

Sara RJ Wilson

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Education

Michigan State University Doctor of Philosophy in Computational Mathematics, Science, and Engineering <i>Advised by Prof. Andrew Christlieb and Prof. Brian O'Shea</i>	East Lansing, MI Expected May 2031
University of Pittsburgh Bachelor of Science in Mathematics <i>Cum Laude</i>	Pittsburgh, PA May 2026

Research Interests

computational fluid dynamics; turbulence; propulsion; scientific machine learning

Honors & Awards

Edna M. Heck Scholarship
Society of Women Engineers Certificate of Merit in Math and Science

Publications

An Analysis of a 2×2 Keyfitz-Kranzer Type Balance System with Varying Generalized Chaplygin Gas <i>Physics of Fluids</i> J. Frew, N. Keyser, E. Kim, G. Paddock, C. Tumbleston, S. Wilson , C. Tsikkou 10.1063/5.0231413 🔗	Sept 2024
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Research Experience

Michigan State University Graduate Student Researcher Current work focuses on instability and failure modes in sparse identification methods (WSINDy) applied to magnetohydrodynamic regimes.	East Lansing, MI Jun 2026 - Current
Semiconductor Research Corporation REU Investigated computational modeling of wireless power transfer systems for 3D integrated devices. Performed electromagnetic simulations to analyze inductance, coupling, and efficiency across coil geometries in coupled physical systems.	Tuscaloosa, AL Jun 2025 – Jul 2025
West Virginia University REU Studied numerical methods for a 2×2 Keyfitz–Kranzer type balance law system with variable Chaplygin gas. Implemented finite difference schemes to analyze Riemann problems with time-dependent wave speeds.	Morgantown, WV Jun 2024 – Jul 2024

Skills

Programming: Python; C++; Java; MATLAB
Computational Methods: Riemann solvers (Lax-Friedrichs/Godunov/HLL); discontinuous Galerkin methods; $h(\text{div})$ -conforming elements
Machine Learning: Ridge; LASSO; STLSQ; SINDy; WSINDy
Tools & Software: Git; $\text{\LaTeX}$; ParaView; Ansys Maxwell

Talks & Presentations

Directed Reading Program, Carnegie Mellon University <i>Computational Plasma Physics</i>	Pittsburgh, PA Dec 2025
Summer Research Symposium, University of Alabama <i>Wireless Power Transfer</i>	Tuscaloosa, AL Jul 2025

Pi Mu Epsilon Poster Session, Joint Mathematics Meetings <i>Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System</i>	Seattle, WA Jan 2025
Summer Research Symposium, West Virginia University <i>Numerical Analysis of the Riemann Problem for a Cosmological 2×2 Balance System</i>	Morgantown, WV Jul 2024

Additional Training

Directed Reading Program, Carnegie Mellon University <i>Computational Plasmas Physics & Nodal Discontinuous Galerkin Methods</i>	Pittsburgh, PA Sept 2025 - Apr 2026
Summer School, Princeton University <i>Fluids and Computer Assisted Proofs</i>	Princeton, NJ Aug 2025
Summer School, Texas A&M University <i>Modeling and Simulation of Partial Differential Equations</i>	College Station, TX May 2025

Teaching Experience

University of Pittsburgh Teaching Assistant <i>Algebra, Business Calculus</i>	Pittsburgh, PA Aug 2024 - Apr 2026
University of Pittsburgh Mathematics Tutor <i>Algebra, Calculus I-III, Linear Algebra, Differential Equations, Theoretical Mathematics, Numerical Analysis</i>	Pittsburgh, PA Jan 2023 - Apr 2026

Leadership & Outreach

University of Pittsburgh Collaborated with the Mathematics Assistance Center advisor to develop and lead accessible tutoring initiatives for nontraditional students with diverse educational backgrounds and needs.	Pittsburgh, PA Jan 2025 - Apr 2026
West Virginia University Mentored incoming REU students in MATLAB, numerical analysis, and scientific computing that contributed to acknowledgments in two subsequent publications.	Remote Jun 2025 - Jul 2025